

1. Administrative & Study Identification

Full Study Title: Non-interventional Study of Ocrelizumab in Participants With Relapsing or Primary Progressive Multiple Sclerosis **Official Title:** A Multicentre Non-interventional Study to Assess the Real-world Effectiveness of Ocrelizumab in Patients With Relapsing or Primary Progressive Multiple Sclerosis - The MuSicalE STUDY **Short Title/Acronym:** MuSicalE Study **Protocol Number:** MN39889 (IRAS ID: 252874) **NCT Number:** NCT03593590 **Posting ID:** 944707831009178376 **Sponsor/Company Name:** Roche (F Hoffmann-La Roche Ltd) **Investigational Product:** Ocrelizumab (Ocrevus) **Phase of Development:** Not Applicable (Observational / Non-interventional) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Head, EU/ROW Regulatory Affairs Pharmaceutical Division, PDR (F Hoffmann-La Roche Ltd)

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the real-world effectiveness of ocrelizumab in patients with relapsing multiple sclerosis (RMS) or primary progressive multiple sclerosis (PPMS) by evaluating the change in the overall SymptoMScreen score over a 4-year period. **Secondary Objectives:** To evaluate the safety profile (percentage of participants with adverse events [AEs], time to treatment discontinuation due to AEs) and changes in other conventional MS and participant-reported outcomes (PROs) reflecting clinical domains such as fatigue, hand function, and cognition, as well as their impact on quality of life and activities of daily living.

2.2 Background & Rationale Current knowledge of ocrelizumab's efficacy and safety stems largely from strictly controlled phase II and phase III clinical trials involving highly selected populations. This study is being conducted to address the need for robust real-world data, observing how ocrelizumab performs when prescribed as part of routine clinical practice in a broader, generalizable population of RMS and PPMS patients.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational Model: Prospective Cohort **Control Type:** Single-arm (Real-world cohort) **Blinding:** Open-label / Non-interventional **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 1,710 participants **Number of Sites:** Multicentre (Global) **Estimated Study Start:** 12-Nov-2018 **Estimated Primary Completion:** 19-Jan-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Must have a definite diagnosis of RMS or PPMS and in whom a decision to initiate ocrelizumab has been taken as part of routine clinical practice. 3. Must be prescribed ocrelizumab in line with the Summary of Product Characteristics (SmPC). 4. Must be able and willing to complete the PROs as per clinical practice.

4.2 Exclusion Criteria 1. Previously treated with ocrelizumab (including phase II, phase III, and phase IIIB clinical trials, investigator-initiated studies, compassionate use, etc.). 2. Not receiving ocrelizumab in line with the approved SmPC.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: Ocrelizumab **Trade Name:** Ocrevus **ATC Code:** L04AA36 **Dosage/Frequency:** Ocrelizumab administered in accordance with approved local labeling / SmPC guidelines for routine practice. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 4 years of observation period per patient, plus an additional 6 months of safety follow-up. **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/ocrevus-epar-product-information_en.pdf

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for symptom management and co-morbidities as per routine clinical practice. **Prohibited:** Any usage of ocrelizumab outside the approved labeling (SmPC parameters) or prior exposure to the drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Clinical Evaluation, Decision to prescribe ocrelizumab
Baseline	Day 0	First Infusion, Baseline PROs (e.g., SymptoMScreen), EDSS Assessment
Interim	Pre-infusions	Routine Care Visits: PRO Completion via BYOD app, Routine MRIs, AE reporting
End of Study	Year 4	Final Efficacy Assessment, Final SymptoMScreen
Safety Follow-Up	Year 4.5	Safety Monitoring, Final AE/SAE reporting 6 months post-study

(Note: Actual visit cadence is aligned with each patient's routine medical care and standard ocrelizumab infusion schedule).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in the overall SymptomScreen score in participants with RMS and PPMS over 4 years. **Measurement Method:** SymptomScreen battery of 7-point Likert scales for 12 distinct functional domains (mobility, dexterity, body pain, sensation, bladder function, fatigue, vision, dizziness, cognition, depression, and anxiety).

7.2 Secondary & Exploratory Endpoints Percentage of RMS and PPMS participants with AEs over 4 years. Time to treatment discontinuation due to adverse events with ocrelizumab. Change in other MS-specific PROs and clinician-measured indices (e.g., EDSS stability).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collected continuously during routine clinical practice interactions; captured comprehensively through study registries. **Serious Adverse Events (SAEs):** Reported as per standard pharmacovigilance regulations for approved commercial therapies. **Data Monitoring Committee (DMC):** Overseen by the Sponsor's internal global safety and regulatory affairs teams.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Real-world Safety and Effectiveness Sets consisting of RMS and PPMS cohorts initiating ocrelizumab. **Sample Size Justification:** Target $N=1,710$ provides sufficient real-world volume to detect statistically significant longitudinal changes in primary outcome PROs across relapsing and progressive sub-cohorts. **Handling of Missing Data:** Standard handling for observational cohorts; missing longitudinal PRO data accounted for using mixed models for repeated measures (MMRM) or multiple imputation depending on missingness mechanisms.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with the Declaration of Helsinki, Good Pharmacovigilance Practices (GPP), and applicable local regulatory requirements for observational studies. **Monitoring Plan:** Source data verification and routine monitoring to ensure accurate transcription from electronic medical records to the study database. **Audit Trail:**

Secure database for PROs collected via digital applications (e.g., BYOD approach) and eCRF forms completed by clinicians.

11. Study Identifiers & Governance

Primary ID: MN39889 **Registry Number:** NCT03593590 **Posting ID:** 944707831009178376 **Sponsor/Lead Organization:** Roche **Study Title:** MuSicalE Study **Official Regulatory Title:** A Multicentre Non-interventional Study to Assess the Real-world Effectiveness of Ocrelizumab in Patients With Relapsing or Primary Progressive Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Disease Area / Preferred Name: Multiple Sclerosis **Semantic Type:** Disease or Syndrome **Preferred UMLS Name:** Ocrelizumab in Multiple Sclerosis **Definition:** A progressive autoimmune disorder affecting the central nervous system resulting in demyelination. Patients develop physical and cognitive impairments that correspond with the affected nerve fibers. **Primary Condition:** Relapsing Multiple Sclerosis (RMS) & Primary Progressive Multiple Sclerosis (PPMS) **MeSH Code:** D009103 **SNOMED ID:** 24700007 **Diagnosis Threshold:** Confirmed RMS or PPMS diagnosis requiring disease-modifying therapy (DMT) **Medical Methodology:** Routine Clinical Care / Approved SmPC Guidelines **Optimization Target:** Stabilization or improvement in functional domains (SymptoMScreen) and EDSS

13. Study Procedures & Methodology

Management Pathway: Standard routine clinical pathway for MS management and IV infusions **Education Protocol (HCP):** Standard SmPC and pharmacovigilance training **Education Protocol (Patient):** Routine counseling prior to infusion; training on "Bring Your Own Device" (BYOD) application for independent PRO submission **Quality Audit Mechanism:** Tracking of PRO compliance prior to treatment cycles and continuous safety registry monitoring

14. Observation & Follow-Up Schedule

Total Duration: 4 years (+ 6-month safety follow-up) **Onsite Visit Cadence:** Aligned with routine ocrelizumab infusion schedules (typically every 6 months) **Remote Monitoring Frequency:** PROs scheduled prior to infusions **Checklist Review Interval:** Reviewed at each clinical visit interval

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					